

3-2-1: Examining IP Addressing Issues

At the end of this episode, I will be able to:

1. Identify and explain the significance of IP addressing issues, given a scenario.

Learner Objective: *Identify and explain the significance of IP addressing issues, given a scenario.*

Description: In this episode, the learner will examine common IP addressing issues in today's TCP/IP networks. We will explore address pool exhaustion, incorrect IP addresses, subnet masks, gateway settings, and more.

- What can cause a 169.254.x.y address?
 - o Address Pool Exhaustion
 - ♣ Occurs when all IP addresses in a designated pool are in use, preventing new device connections
 - ♣ Common issues
 - ♣ High number of connected devices exceeding the available addresses in the DHCP scope
 - ♣ Solutions
 - ♣ Expand the DHCP address pool or reduce lease time to free up addresses
 - o Duplicate IP Address
 - ♣ Two hosts are assigned the same IP address on a network
 - ♣ Common issues
 - ♣ Manual configuration errors
 - ♣ DHCP misconfigurations
 - ♣ Solutions
 - ♣ Use DHCP reservation for critical devices
 - ♣ Verify static IPs do not overlap with the DHCP scope
- Could this be from misconfigured network adapter settings?
 - o Incorrect Default Gateway
 - ♣ The designated router IP for outgoing traffic is wrongly configured, hindering external network access
 - ♣ Solutions
 - ♣ Verify and correct the default gateway settings on affected devices
 - ♣ Reconfigure and update DHCP server settings
 - o Incorrect IP Address
 - ♣ An IP address that does not align with the intended network configuration is assigned to a device
 - ♣ Solutions
 - ♣ Ensure static IP addresses are correctly assigned
 - ♣ Reconfigure and update DHCP server settings
 - o Incorrect Subnet Mask
 - ♣ An improperly configured subnet mask can lead to incorrect network or broadcast addresses on devices
 - ♣ Solutions
 - ♣ Double-check and correct subnet mask configurations on individual devices
 - ♣ Reconfigure and update DHCP server settings*
- You mentioned DHCP scope options, what are they?
 - o Configurable settings within a DHCP server
 - o Provide clients with additional parameters:
 - ♣

- - ♣ Default gateway
 - ♣ DNS servers
 - ♣ Subnet mask
 - These setting can be manually or setting dynamically?
 - o Manually which is also called static, as the setting do not change
 - ♣ Not scalable
 - ♣ Prone to human error
 - o Dynamically, through the DHCP service
 - ♣ Greater scability
 - ♣ Less prone to human error
 - ♣ Service outages and reconfigurations can cause issues
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- Additional Resources:
 - o If applicable
 - o Common issues
 - ♣ Static or manual configuration errors
 - ♣ DHCP misconfigurations